Global Fuel Prices and Taxation Analysis (2020–2026)

Project Report: Data Analytics with Tableau

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1. INTRODUCTION

1.1 Project Overview

The international energy market is undergoing significant structural shifts driven by geopolitical events, carbon taxation initiatives, and varying national fuel subsidy policies. This project analyzes global fuel market records across 84 countries, 7 world regions, 4 income tiers, and 27,468 records (2020–2026), tracking Petrol, Diesel, LPG, national tax burdens, subsidy schemes, and Brent crude spot prices through two interactive Tableau dashboards and a guided data story.

- **Dashboard 1 (Policy & Tax Structure View):** Top KPI cards (84 countries, $1.42/L avg petrol price, 41.8% avg tax share, $95.4B net subsidies), geographic tax heatmaps, subsidy volume packed bubbles, and fuel tax percentage distributions.
- **Dashboard 2 (Logistics & Econometric Research View):** Multi-year price trends (2020–2026), Brent crude spot price pass-through correlation models, income-level fuel affordability cross-tabs, and regional carbon offset impact.

Tableau Public Link: https://public.tableau.com/views/GlobalFuelPricesAndTaxationAnalysis2020-2026/Dashboard1?:language=en-US&:display_count=n&:origin=viz_share_link

1.2 Purpose

The purpose is to transform raw, scattered energy commodity records into actionable Business Intelligence visual dashboards that assist policy consultants, freight supply chain managers, and energy economists in evaluating tax policies, predicting logistics risks, and modeling crude price transmission.

2. IDEATION PHASE

2.1 Problem Statement & Personas

Global fuel pricing data is large and fragmented across international reports. Decision-makers lack a single unified view to monitor fiscal taxation, freight fuel volatility, and crude pass-through effects. Three core stakeholders require tailored analytics:

Persona	Role	Need
Dr. Sarah Mitchell	Policy Consultant	High-tax vs. subsidized market benchmarking, fiscal tax burden heatmaps, and carbon policy comparison.
Marcus Thompson	Supply Chain Manager	Multi-year diesel price trends, geographic transit corridor volatility mapping, and price forecasting.
Prof. Elena Rodriguez	Energy Economist	Brent crude pass-through correlation modeling, income-level affordability cross-tabs, and structural comparison.

2.2 Empathy Map Canvas

Quadrant	Dr. Sarah Mitchell (Policy)	Marcus Thompson (Logistics)	Prof. Elena Rodriguez (Research)
Says	"Which countries levy the highest tax burden?"	"Where are fuel price spikes impacting routes?"	"How strongly does crude correlate with pump rates?"
Thinks / Does	Needs credible evidence for fiscal reform proposals.	Monitors diesel pricing to hedge logistics budgets.	Evaluates cross-country income level affordability.
Feels	Pressure to balance fiscal revenue with energy access.	Anxious over unbudgeted freight fuel cost surges.	Motivated by rigorous economic policy evidence.

2.3 Brainstorming & Solution Framing

Idea groupings selected for visual implementation in Tableau:

- **Policy & Fiscal:** Tax percentage heatmaps, net subsidy packed bubbles, tax vs. price scatter plots.
- **Logistics & Supply Chain:** Diesel price distribution maps, 2020–2026 multi-fuel trendlines, regional volatility index.
- **Econometrics & Research:** Brent crude correlation dual-axis charts, income tier affordability cross-tabulations.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map Overview

Mapped stakeholder interactions across 5 key phases: **Awareness** (KPI tiles review), **Exploration** (Region & fuel filtering), **Analysis** (Scatter plots & box plot deep-dives), **Decision** (Strategy selection), and **Sharing** (Exporting PDF briefs).

3.2 Solution Requirements

Functional Requirements: Display global KPIs, map geographical tax distributions, compare multi-fuel price trends (2020–2026), compute Brent crude pass-through correlations, and provide interactive quick-filters.

Non-functional Requirements: Optimized workbook performance (< 0.8s query time), consistent color theming, complete data validation across 27,468 rows, and public availability on Tableau Server.

3.3 Data Flow Diagram & Tech Stack

Data Pipeline: Raw Global Fuel CSV → Tableau Extract Engine (.hyper) → Calculated Fields & Aggregations → 12 Visual Worksheets → 2 Scenario Dashboards → Tableau Story → Executive Delivery.

4. PROJECT DESIGN

4.1 Problem Solution Fit

The problem is large, scattered fuel market data that prevents swift policy and supply chain decision-making. The solution provides a unified 2-dashboard suite in Tableau, replacing hours of manual spreadsheet processing with instant visual insights.

4.2 Proposed Solution Layout

- **Dashboard 1 (Policy & Tax Overview):** Global KPI tiles, tax burden heatmaps, packed bubble subsidy charts, tax % distribution.
- **Dashboard 2 (Logistics & Econometrics):** Multi-year trendlines, Brent crude spot price correlation, income tier cross-tabs, predictive volatility forecasting.
- **Story Points:** 5-step narrative guiding stakeholders through key project discoveries.

4.3 Solution Architecture

A structured 7-layer architecture: Data Source Layer (CSV) → Data Connection → Data Preparation (Currency Normalization) → Data Layer (.hyper Extract) → Analytics Layer (LOD Expressions & Measures) → Visualization Layer → End-User Scenario Layer.

5. PROJECT PLANNING & SCHEDULING

5.1 Sprint Execution & Burndown Summary

The project was executed across 3 planned Agile sprints totaling **53 story points** over 6 days (20 Oct – 25 Oct 2026). The team achieved an Average Velocity of 17.67 story points per sprint (8.83 points per day), burning down all backlog items to 0 by the target completion date.

Sprint	Points	Start Date	End Date	Status
Sprint-1	20	20 Oct 2026	21 Oct 2026	Completed
Sprint-2	20	22 Oct 2026	23 Oct 2026	Completed
Sprint-3	13	24 Oct 2026	25 Oct 2026	Completed

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Test Execution Results

Test Case	Expected Result	Status
KPI Cards Verification	Displays 84 countries, $1.42/L avg price, 41.8% tax share, $95.4B subsidies accurately.	**Pass**
Global Quick-Filter Sync	Selecting Region or Fuel Type dynamically updates all charts across both dashboards.	**Pass**
Calculated Fields Logic	LOD expressions compute Net Tax Share % and Volatility Index without mathematical error.	**Pass**
Tableau Server Render Test	Dashboards render fully in web browsers within < 0.8 seconds.	**Pass**
Story Point Navigation	All 5 story points transition seamlessly without data extract disconnects.	**Pass**

7. RESULTS & FINDINGS

7.1 Key Insights Discovered

- **Tax Burden Concentration:** European nations impose the highest average tax percentage on retail petrol (up to 62%), contrasting with North American markets (18–25%).
- **Subsidy Distortion:** Net fuel subsidies are heavily concentrated in Middle Eastern and North African economies ($95.4B combined), keeping consumer prices artificially low.
- **Freight Volatility:** Diesel pricing across South Asian freight transit corridors exhibits the highest 12-month volatility index, presenting budget risks for logistics operators.
- **Crude Oil Pass-Through:** Brent crude spot price fluctuations transmit to retail petrol pump rates with a 3-to-4 week lag, showing a strong positive correlation (r = 0.84).
- **Income Tier Disparity:** Low-income nations experience higher relative fuel price volatility, spending up to 18% of per-capita GDP on transport energy compared to 4% in high-income economies.

8. ADVANTAGES & DISADVANTAGES

8.1 Solution Advantages

- Provides a single interactive view combining fiscal, supply chain, and macroeconomic fuel data.
- Empowers 3 distinct stakeholder personas with custom visual scenarios within one unified workbook.
- Enables rapid web sharing and executive reporting via Tableau Public.
- Drives data-backed decision-making for carbon tax policy formulation and freight route hedging.

8.2 Project Limitations

- Relies on historical batch dataset extracts (2020–2026) rather than real-time live exchange streams.
- Public Tableau hosting makes aggregated market figures publicly visible.
- High data density across 84 countries requires quick-filter selection for optimal mobile viewing.

9. CONCLUSION

This project demonstrates how Tableau Desktop can successfully convert massive, fragmented global energy commodity data into clear, actionable Business Intelligence insights. The delivered dashboards reveal critical tax policy differences, logistics cost vulnerabilities, and crude pass-through dynamics across 84 countries—providing policy consultants, supply chain directors, and energy economists with a robust analytical foundation.

10. FUTURE SCOPE

- Connect live streaming API feeds from global commodity energy exchanges.
- Integrate 24-month machine learning predictive price forecasting models.
- Incorporate electric vehicle tariff comparison layers alongside fossil fuel benchmarks.
- Build mobile-optimized dashboard layouts for smartphone access.

11. APPENDIX & PROJECT LINKS

11.1 Project Deliverable Repositories

Tableau Public Live Story & Dashboard Link:
https://public.tableau.com/views/GlobalFuelPricesAndTaxationAnalysis2020-2026/Story1?:language=en-US&:display_count=n&:origin=viz_share_link

GitHub Project Repository:
https://github.com/DivyanshiJain/GlobalFuelPricesAndTaxationAnalysis

Final Project Report Sign-Off:

Prepared By: Divyanshi Jain **Verified By:** Course Coordinator **Approved Date:** 31 October 2026